

Muhammad Hamid Raza

Software Engineer | Full-Stack & ML Developer

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PROFESSIONAL SUMMARY

Full-stack developer and machine learning engineer with hands-on experience building and deploying AI-powered web applications. Skilled in Python, React, and FastAPI with proven ability to deliver production-ready systems from model training through cloud deployment. Experienced in real-world data science through a remote internship and multiple deployed projects.

EXPERIENCE

Data Scientist Intern — DevelopersHub Corporation

Jul 2025 – Sep 2025 · Remote, Islamabad, Pakistan

- Built and compared 5 classification models (Logistic Regression, KNN, Decision Tree, SVM, Random Forest) across Iris, Credit Risk, and Customer Churn datasets using Scikit-learn
- Developed a bank term deposit prediction model using Random Forest with SHAP-based feature explainability and ROC/F1 evaluation metrics
- Built a K-Means customer segmentation pipeline with PCA and t-SNE visualization, and an interactive Streamlit + Plotly business dashboard for global superstore sales analytics

EDUCATION

University of Agriculture, Faisalabad

Bachelor of Science in Computer Science

Aug 2022 – May 2026

Core Coursework: Data Structures & Algorithms, OOP, Operating Systems, Database Systems, Computer Networks, Software Engineering, AI, Machine Learning, Web Technologies, Digital Signal Processing

Punjab Group of Colleges

FSC Pre-Medical — Grade: 94.7% (Top Position)

Jun 2020 – May 2022

TECHNICAL SKILLS

- Programming Languages** — Python, JavaScript, TypeScript, C, C++
- Web Development** — React, FastAPI, Flask, Vite, Tailwind CSS, HTML5, CSS3, Bootstrap, WordPress
- Cloud & Deployment** — Vercel, Streamlit Cloud, Azure App Service, Docker, Railway
- Version Control** — Git, GitHub
- Databases** — MongoDB, PostgreSQL, MS SQL, SQL
- Data Science** — Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Tableau, Plotly
- Machine Learning** — Supervised Learning, Unsupervised Learning, Model Deployment, PyTorch, TensorFlow, Keras
- Tools** — Jupyter Notebook, Anaconda, Google Colab, VS Code
- Operating Systems** — Windows, Linux (basic operations)

FINAL YEAR PROJECT

Full Stack Developer — MediScan AI | mediscanai.me | 2025–2026

- Built a 26-class histopathology classifier (MobileNetV2), 4-class brain MRI classifier (CNN), and biomedical NLP report analyser (BioBERT NER) — covering 30+ disease categories across 3 AI models
- FastAPI + PyTorch backend deployed on Azure App Service (Docker), React + Vite frontend on Vercel with CI/CD via GitHub Actions
- Implemented two-phase transfer learning, confidence thresholding, and binary medical image validator (EfficientNet-B0 + CLIP fallback)

PERSONAL PROJECTS

UAF Result Hub | uafresult.live | 2025

- Built with React 18, TypeScript, Vite, Tailwind CSS, Shadcn UI — deployed on Vercel, serving UAF students across all departments
- Vercel serverless proxy automating UAF LMS login (CSRF + session) and parsing HTML results with Cheerio
- GPA/CGPA analytics dashboard, target CGPA calculator, smart search, and class result comparison across 5+ analytical features

Video Downloader | video-downloader-jade.vercel.app | 2025

- FastAPI + yt-dlp + FFmpeg backend (Railway), React + Vite + Tailwind CSS frontend (Vercel)
- Multi-platform support across 4 platforms: YouTube, Facebook, Instagram, TikTok — quality selection and MP3 extraction
- Rate limiting, auto file cleanup, URL validation, and platform detection for production reliability

Mikel Saez de Vicuña— Portfolio Website | saezdevicuna.eus | 2026

- Built full-stack portfolio for Spain-based Upwork client using React 19 + Vite, from scratch to production deployment on Netlify
- Implemented multilingual support (English, Spanish, Basque) via Google Translate API, GSAP scroll animations, and custom cursor trail effects
- Integrated interactive world map (Leaflet + GeoJSON) with dynamic pins fetched live from Google Sheets via PapaParse
- Developed custom CMS with email-based authentication, enabling client to update website content without touching code

HR Attrition Analysis — ML Classification | 2024

- Applied 3 classification models (Logistic Regression, Decision Tree, Random Forest) on HR dataset of 14,999 employee records
- Evaluated results using confusion matrix metrics: precision, accuracy, recall and F1 — Random Forest achieved highest accuracy among all 3 models
- Built a Streamlit web interface for interactive result visualization

HACKATHONS & COMPETITIONS

- **IAAC Final Round — Silver Honour:** Scored 16/20, ranked in top 8% among all global participants
- **Calico Fall '24 Hackathon:** Participated in timed problem-solving competition tackling real-world challenges
- **NASA Space Apps Challenge 2024:** Collaborated on "Galactic Problem Solver" project, presented prototype at global hackathon
- **Defense Day Poster Design Competition, UAF:** Created commemorative poster using Photoshop; awarded competition certificate

CERTIFICATIONS & COURSES

- **Learn to Code in Python 3** — Udemy (Ivan Lourenço Gomes) | 2023
- **Introduction to Front-End Development** — Coursera (META) | 2023
- **Programming for Everybody (Python)** — Coursera (University of Michigan) | 2023
- **Foundations: Data, Data, Everywhere** — Coursera (Google) | 2024
- **Build a Computer Vision App with Azure Cognitive Services** — Coursera (Microsoft) | 2024